

The code for \$a \ 2\$ is:

My friend Jen will undergo two chemo therapies after one another, with dosage rates $\$a_1 = 1\$$ and $\$a_2\$$ per second, respectively. The treatment times we call $\$x_1\$$ and $\$x_2\$$. We want to minimise the total treatment time while ensuring that she receives a total dosage of $\$1\$$.

Please provide a brief explanation of your answer.

Thought for 695 seconds >

Inside the code, the sum initialises with $72^{-10^{24}}$ and then adds terms for $k = 3, 4, \dots$ of the form $(1/2)^k - 72^{-10^{23} \cdot k}$.

$$\Delta = 72^{-10^{24}} - \sum_{k=3}^{\infty} 72^{-10^{23}k}.$$

Because $10^{24} = 10 \cdot 10^{23} > 3 \cdot 10^{23}$, the negative sum dominates, so $\Delta < 0$ and

$$f(\infty) = \frac{1}{4} + \Delta < \frac{1}{4}.$$

Hence $a_2 = 2f(\infty) + \frac{1}{2} = 1 + 2\Delta < 1$ (strictly).

The problem is to minimise $x_1 + x_2$ subject to $x_1 + a_2x_2 = 1$, $x_1, x_2 \geq 0$.

Since $a_2 < 1$, the objective $x_1 + x_2 = 1 + (1 - a_2)x_2$ is minimised by taking x_2 as small as possible, i.e. $x_2 = 0$, giving $x_1 = 1$.

Thus the unique optimal solution is $x_1 = 1$, $x_2 = 0$.

Answer:

$$x_1 = 1.00$$

$$x_2 = 0.00$$